Cloud Operations for ExaDB-D

Provisioning Operations Scenarios

Operations Advisory Team

Office of CTO | EMEA Solution Engineering

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Agenda

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- Provisioning Operations Scenarios
  - Provision Exadata Infrastructure
  - Provision VM Cluster Networks
  - Provision VM Clusters
  - Provision Oracle Homes
  - Provision Container Databases (CDBs)
  - Provision Pluggable Databases (PDBs)
  - Provision Data Guard Association

Introduction

Introduction

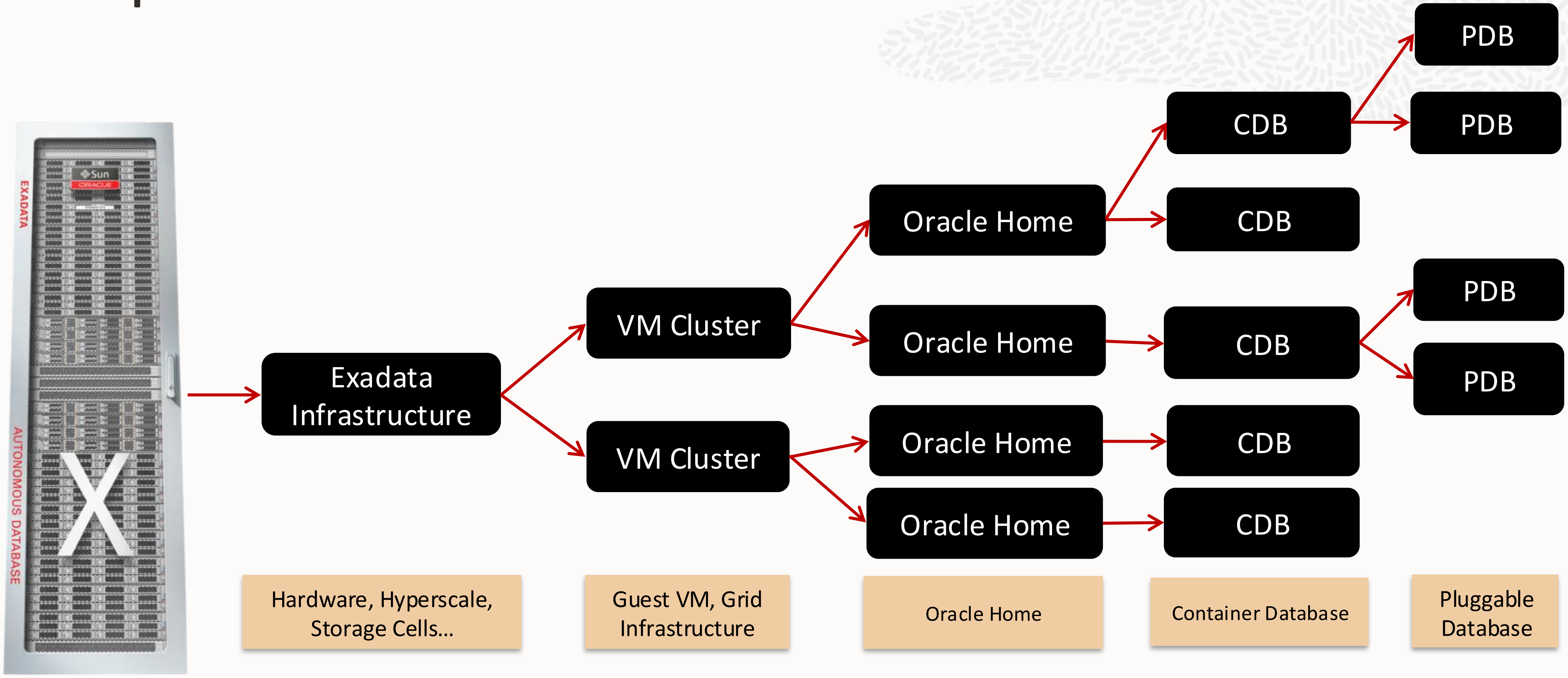
This Workshop provides information about Exadata Database Service on Dedicated Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

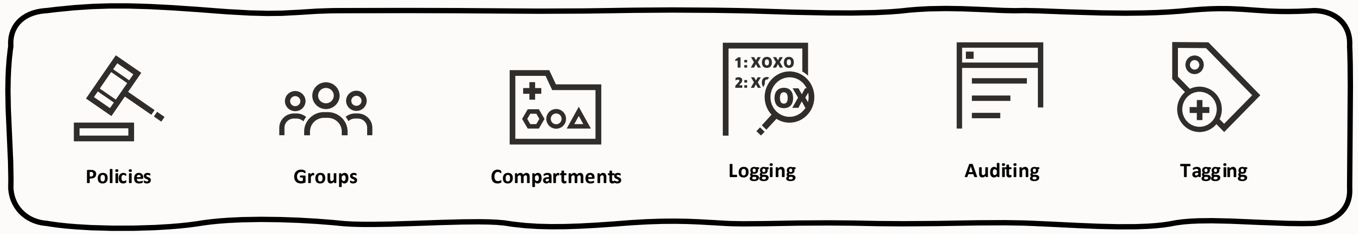
We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VMC, OH, DBs, DG, scaling and patching

The code in this Workshop is just sample code and can be incomplete.

To make the sample code work in your Oracle Cloud tenancy,
please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

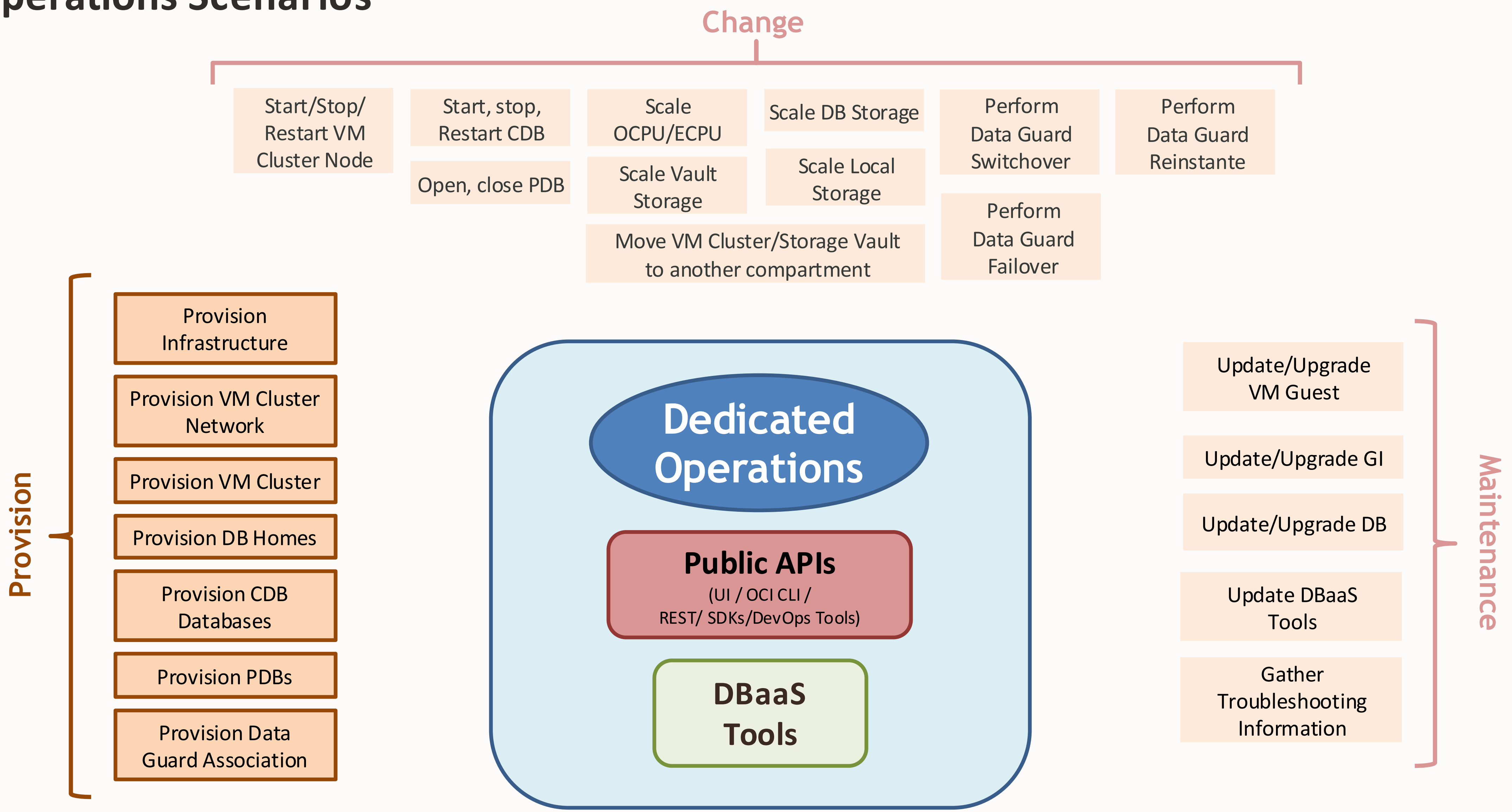
Main Components





Provision Operations

Operations Scenarios



Provision Infrastucture

Provision an Exadata Infrastructure

An Exadata Infrastructure represents the dedicated Exadata rack that provides the compute, storage, and networking resources used to host VM Clusters.

Requirements

- OCI Tenancy with required IAM permissions
- Availability Domain selected
- Exadata Infrastructure service limit available

Compute and Storage Configuration

- If you select the flexible cloud infrastructure model, your initial Infrastructure instance can have a **minimum of 2 database servers and 3 storage servers** up to 32 database servers and 64 storage servers.
- After provisioning, you can scale the service instance as needed by adding additional storage servers, compute servers, or both.
- You will see the Resource Allocation per server.

Maintenance Preferences

- Configure the preferred maintenance window
- Select maintenance method.
- Specify customer notification contacts.

How to use Storage Vault (if possible)

- Create a Storage Vault after provisioning the Exadata Infrastructure
- Define the vault name and availability domain
- Allocate the required storage capacity
- The Storage Vault is used by Exascale VM Clusters

How can you create an Exadata Infrastructure?

You can create an Exadata Infrastructure by using:

- OCI Console, OCI-CLI, REST API, SDK, IaC

OCI Control Plane Interfaces

OCI Console

Infrastructure *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata Infrastructure ➔* ***Create Exadata Infrastructure***

Storage Vault *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata Infrastructure ➔ Select the Infrastructure ➔ Exascale Storage Vaults ➔* ***Create Exascale Storage Vault***

Command Line Interface (OCI CLI)

Create an Exadata Infrastructure

```
oci db cloud-exa-infra create \
  --compartment-id <compartment_ocid> \
  --availability-domain <av_dom_ocid> [OPTIONS]
```

Create Storage Vault

```
oci db cloud-exa-infra configure-exascale \
  --cloud-exa-infra-id <exa_infra_ocid> \
  --total-storage-in-gbs [integer] [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_cloud_exadata_infrastructure"
"test_exa_infra" {
    #Required
    availability_domain =
var.cloud_exadata_infrastructure_availability_domain
    compartment_id = var.compartment_id
    display_name =
var.cloud_exadata_infrastructure_display_name
    shape = var.cloud_exadata_infrastructure_shape
…
}

resource "oci_database_exascale_db_storage_vault"
"test_exascale_db_storage_vault" {
    availability_domain = var.exascale_db_sto_vault_AD
    compartment_id = var.compartment_id
    exadata_infrastructure_id =
oci_database_cloud_exadata_infrastructure.test_exa_infra.id
…
}
```

Provision VM Cluster Networks

Provision VM Cluster Network

Requirements

- A **VCN** in the region
- At least **two subnets** in the VCN:
  - Client subnet
  - Backup subnet
- Choose which method of DNS name resolution you will use

Recommendations

- Use regional subnets (span all availability domains in the region)
- Custom route tables for each subnet
- Create security rules to control traffic to and from the client network and backup network of the VM Cluster nodes

Options

- Option 1: Public Client Subnet with Internet Gateway
- Option 2: Private Subnets (recommended for production)

Requirements for IP Address Space

- IP addresses must not overlap
- Client subnet:
  - Requires **4** IP addresses **per node**
  - **3** addresses are reserved for SCANs
- Backup subnet
  - Requires **3** IP addresses **per node**
- The Networking service reserves **3** IP addresses in **each subnet**
- To plan for future growth, add addresses that you expect to require as you scale up your VM Cluster, not only the number of VMs you plan to provision for immediate need
- **100.64.0.0/10 address space is reserved for the interconnect**

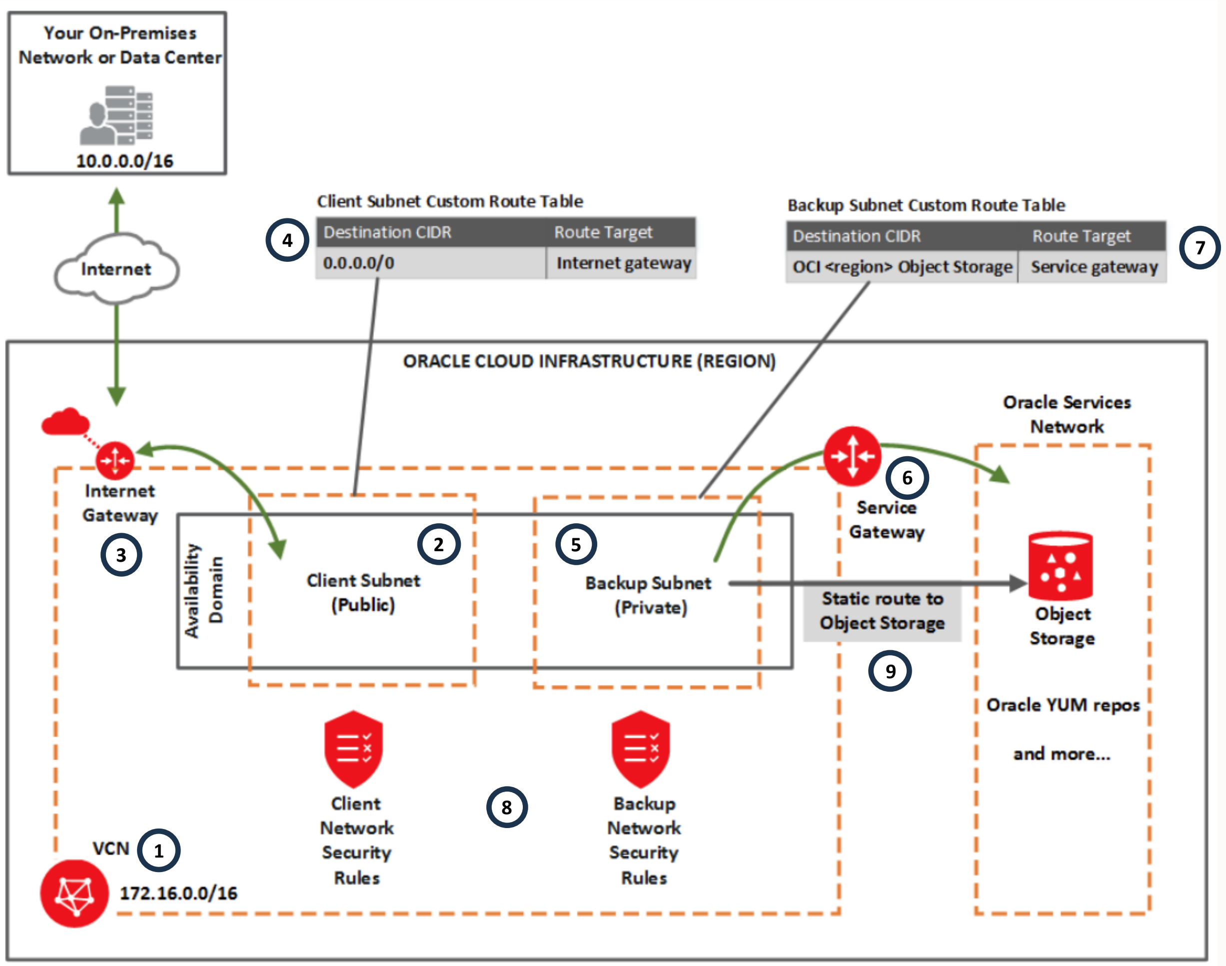
How can you create network components?

You can create network components by using:

- OCI Console, OCI CLI, REST API, SDK, IaC

Network Setup

Option 1: Public Client Subnet with Internet Gateway



6 7 8

ONLY OBJECT STORAGE ACCESS

You configure the backup subnet to use the service gateway for access only to Object Storage.

9

FOR CUSTOMIZED BACKUPS

All the traffic is, by default, routed through the data network. To route backup traffic to the backup interface (BONDETH1), you need to configure a static route on each of the compute nodes in the cluster.

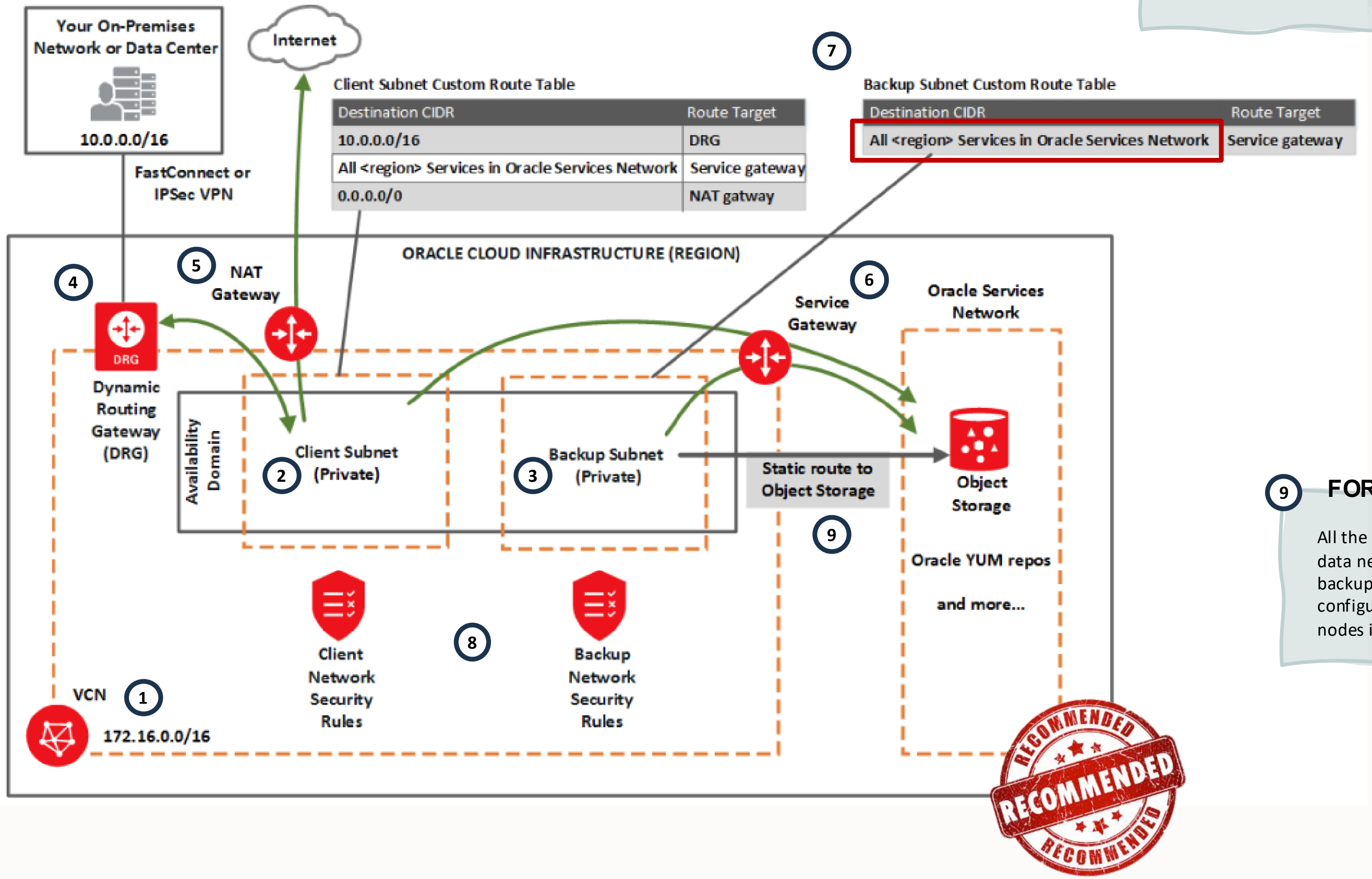
Network Setup

Option 2: Private Subnets

6 **OBJ. STORAGE AND YUM REPOS ACCESS**

7 All the traffic is, by default, routed through the data network. To route backup traffic to the backup interface (BONDETH1), you

8 need to configure a static route on each of the compute nodes in the cluster.



9 **FOR CUSTOMIZED BACKUPS**

All the traffic is, by default, routed through the data network. To route backup traffic to the backup interface (BONDETH1), you need to configure a static route on each of the compute nodes in the cluster.

Network Setup

Security Rules for the Oracle Exadata Database Service on Dedicated Infrastructure

Ingress Rule#	Description	Stateless	Source Type	Source CIDR	IP Protocol	Source Port	Destination Port	Type	Code
General #1	Allows SSH traffic from anywhere	No	CIDR	0.0.0.0/0	SSH	All	22		
General #2	Allows Path MTU Discovery fragmentation messages	No	CIDR	0.0.0.0/0	ICMP			3	4
General #3	Allows connectivity error messages within the VCN	No	CIDR	Your VCN's CIDR	ICMP			3	All
Client #1	Allows ONS and FAN traffic from within the client subnet	No	CIDR	Client subnet's CIDR	TCP	All	6200		
Client #2	Allows SQL*NET traffic from within the client subnet	No	CIDR	Client subnet's CIDR	TCP	All	1521		
Client #3	Allows Patching Traffic from Within the Client Subnet	No	CIDR	Client subnet's CIDR	TCP	All	7085		

Egress Rule#	Description	Stateless	Destination Type	Destination CIDR	IP Protocol	Source Port	Destination Port	Destination Service
General #1	Allows all egress traffic	No	CIDR	0.0.0.0/0	All			
Client #1	Allows all TCP traffic inside the client subnet	No	CIDR	0.0.0.0/0	TCP	All	22	
Client #2	Allows all egress traffic (allows connections to the Oracle YUM repos)	No	CIDR	0.0.0.0/0	All			
Backup #1	Allows access to Object Storage	No	Service		TCP	All	443 (HTTPS)	OCI <region> Object Storage All <region> Services (YUM repo access)

OCI Control Plane Interfaces

OCI Console

Create	*Menu ➔ Networking ➔ Virtual Cloud Network ➔ **Create VCN** or **Actions (Start VCN Wizard)***
Delete	*Menu ➔ Networking ➔ Virtual Cloud Network ➔ Select Actions (3 dots) on the VCN ➔ **Delete***

Command Line Interface (OCI CLI)

Create VCN & Subnet

```
oci network vcn create --compartment-id <compartment_ocid> \
   --cidr-block <cidr_block> [OPTIONS]

oci network subnet create --compartment-id <compartment_ocid> \
   --vcn-id <vcn_id> [OPTIONS]
```

Delete VCN & Subnet

```
oci network vcn delete --vcn-id <vcn-id> [OPTIONS]

oci network subnet delete --subnet-id <subnet-id> [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_core_vcn" "test_vcn" {
    compartment_id = var.compartment_id
    cidr_block = var.vcn_cidr_block
    display_name = var.vcn_display_name
    dns_label = var.vcn_dns_label
...
}

resource "oci_core_subnet" "test_subnet" {
    compartment_id = var.compartment_id
    vcn_id = oci_core_vcn.test_vcn.id
    ipv4cidr_blocks = var.subnet_ipv4cidr_blocks
    route_table_id =
oci_core_route_table.test_route_table.id
    security_list_ids = var.subnet_security_list_ids
...
}
```

Provision VM Cluster

Provision a VM Cluster

*A **VM Cluster** is a group of virtual machines configured to act as a unified environment for hosting Oracle Database instances.*

Requirements

- Cloud Exadata Infrastructure provisioned
- Network configuration: VCN, Subnets and DNS
- SSH Public Key

VM Cluster Types

ASM VM Cluster

- Traditional Exadata deployment model
- Uses ASM disk groups on Exadata storage servers

Exascale VM Cluster

- Uses Exascale Storage Vaults
- Enables Exascale storage architecture on dedicated infrastructure

Provisioning Parameters

Compute Configuration

- Number of VM cluster nodes
- OCPU/ECPU allocation per VM
- Exadata image

Storage Configuration

- VM files and DB Storage
- Storage size allocation

Oracle Grid Infrastructure version

Network Configuration

Access Configuration

The VM Cluster storage architecture is selected during provisioning and cannot be changed after deployment. To use a different architecture, provision a new VM Cluster.

How can you create a VM Cluster?

You can create a VM cluster by using:

- OCI Console, OCI-CLI, REST API, SDK, IaC

Why create additional VM Clusters?

Security separation

- Provide separation between Dev and Test environments, or between multiple production databases
- VMs can be used to address isolation for data security requirements including PCI data and HIPAA data

Maintenance

- Isolate group of database and minimize the potential downtime for GI patching

Blast radius

- Virtualization limits the blast radius in case of system failure

Resource management

- Ensures that database receives the resources it requires
- Minimizes the ‘noisy neighbor’ issue
- Possible licensing limitation approach

Administrative separation

- An organization that have multiple system administrator teams

- Each VM brings additional administrative overhead, including installation, patching, security, and management
- Excessive isolation, such as the “one DB per VM” approach, results in virtual sprawl.
- Oracle recommends a more judicious use of VMs, including deployment of small numbers of VMs per physical machine, with each VM hosting multiple CDBs, and/or PDBs

OCI Control Plane Interfaces

OCI Console

Create — *Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure → Exadata VM Clusters →* ***Create VM Cluster***

Delete — *Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure → Exadata VM Clusters → Select Actions (3 dots) on the VM Cluster →* ***Terminate***

Command Line Interface (OCI CLI)

Create VM Cluster

```
oci db cloud-vm-cluster create --compartment-id <comp_ocid> \
  --cloud-exa-infra-id <exace_infra_ocid> \
  --subnet-id <client_subnet_ocid> \
  --backup-subnet-id <backup_subnet_ocid> \
  --display-name <cluster_name> [OPTIONS]
```

Delete VM Cluster

```
oci db cloud-vm-cluster delete \
  --cloud-vm-cluster-id <vm_cluster_ocid> [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_cloud_vm_cluster" "test_exadb_vm_cl"
{
     backup_subnet_id = oci_core_subnet.test_subnet.id
    cloud_exadata_infrastructure_id =
oci_database_cloud_exadata_infrastructure.test_cloud_exadata
_infrastructure.id
    compartment_id = var.compartment_id
    cpu_core_count = var.cloud_vm_cluster_cpu_core_count
    display_name = var.cloud_vm_cluster_display_name
    gi_version = var.cloud_vm_cluster_gi_version
    hostname = var.cloud_vm_cluster_hostname
    ssh_public_keys = var.cloud_vm_cluster_ssh_public_keys
    subnet_id = oci_core_subnet.test_subnet.id
...
}
```

Provision Oracle Database Homes

Provision an Oracle Database Home

*A **Database Home** is a directory location on the Exadata database virtual machines that contains Oracle Database software binary files.*

How can you create a Database Home?

You can add Oracle Database homes to an existing VM cluster by using:

- Control Plane: OCI Console, OCI CLI, REST API, SDK, IaC
- Guest VM: dbaascli

Database Home Options:

- Oracle-provided Database Home
- Custom Database Software Image
  - If custom one-offs or RUs are required, they can be applied after Database Home deployment; however, Oracle recommends including them in a Custom Database Software Image for repeatable and consistent deployments.

Deployment Options:

- Only deploy a Database Home
- Create a Database Home at Database creation

Characteristics

- Multiple Database Homes per VM Cluster
- Multiple Databases per Database Home
- Independent Database Home lifecycle

Do not download RDBMS binaries from the internet or MOS to install it, as you would be outside automation, probably you would lack of some important patches or one-offs specific for Exadata Cloud and it could be a source of problems

OCI Control Plane Interfaces

OCI Console

Create *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata VM Clusters ➔ Database homes ➔* ***Create database database***

Delete *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata VM Clusters ➔ Database homes ➔ Select Actions (3 dots) on the database home ➔* ***Delete***

Command Line Interface (OCI CLI)

Create database

```
oci db db-home create --vm-cluster-id <vm-cl-ocid> \
  --database-software-image-id <image_ocid> [OPTIONS]
```

Delete database

```
oci db db-home delete --db-home-id <dbhome_ocid> [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_db_home" "test_db_home" {
    database_software_image_id =
oci_database_database_software_image.test_database_software_
image.id
    db_system_id = oci_database_db_system.test_db_system.id
    defined_tags = var.db_home_defined_tags
    display_name = var.db_home_display_name
    vm_cluster_id =
oci_database_vm_cluster.test_vm_cluster.id
           ...
}
```

Local VM Command-Line Interfaces - dbaascli

Database software images are resources within your tenancy that you create prior to provisioning or patching a Grid Infrastructure, a Database Home, or a database.

List locally available DB/GI images

```
$ dbaascli cswLib listLocal [--product DATABASE | GRID]
```

Download/update an image

```
$ dbaascli cswLib download {--imageTag <value> | --version <value> } [--product DATABASE | GRID]
```

List available DB/GI images in CPS

```
$ dbaascli cswLib showImages [--product DATABASE | GRID] [--jsonFormat [--jsonFilePath <value>]]
```

Delete a local image

```
$ dbaascli cswLib deleteLocal --imageTag <value>
```

[1]… Not all parameters listed. Please check the command definition

Local VM Command-Line Interfaces - dbaascli

Create a new database home

```
$ dbaascli dbHome create --version <value> [--oracleHomeName <value>] [--oracleHome <value>] [--imageTag <value>]
  [--enableUnifiedAuditing] [--executePrereqs] [--imageFilePath <value>] [--waitForCompletion <value>]
```

Delete the given database home

```
$ dbaascli dbHome delete {--oracleHome <value> | --oracleHomeName <value>} [--resume [--sessionID <value>]]
  [--waitForCompletion <value>]
```

Display details about a specific database home

```
$ dbaascli dbHome getDetails {--oracleHomeName <value> | --oracleHome <value>}
```

Display details of all the database homes

```
$ dbaascli system getDBHomes
```

Display details of databases in specific database home

```
$ dbaascli dbHome getDatabases {--oracleHomeName <value> | --oracleHome <value>}
```

[1]… Not all parameters listed. Please check the command definition

Provision Container Databases

Provision an Oracle Container Database

*A **Container Database (CDB)** is an Oracle database that functions as a container for zero, one, or many pluggable databases (PDBs).*

Requirements

- VM Cluster Configured
- Backup Destinations Created
  - Autonomous Recovery Service or Object Storage bucket
- TDE Encryption Required
- OH (existing or created during CDB provisioning)

Considerations

- By default, uses Enterprise Edition - Extreme Performance
- CDB created with optional first pluggable database
- Database would be created in all the nodes of the VM Cluster
- SGA_TARGET is set by the automation:
  - VM memory <= 60 GB → SGA_TARGET = 3800 MB
  - VM memory >= 60 GB → SGA_TARGET = 7600 MB.
- All DBs are configured using USE_LARGE_PAGES=ONLY
- Default (and recommended) hugepages space in the OS is equal to half of the total amount of RAM memory

Versions

ASM VM Cluster

- Oracle Database 19c / 26ai
- Uses ASM disk groups
- DATA / RECO disk groups

Exascale VM Cluster

- Oracle AI Database 26ai
- Uses Exascale Storage Vaults
- Direct Storage Vault access

How can you create a Container Database?

You can create an Oracle Container Database to an existing VM cluster by using:

- OCI Console, OCI CLI, REST API, SDK, IaC
- dbaascli

OCI Control Plane Interfaces

OCI Console

Create *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata VM Clusters ➔ Container databases ➔ **Create container database***

Delete *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata VM Clusters ➔ Container databases ➔ Select Actions (3 dots) on the CDB ➔ **Terminate***

Command Line Interface (OCI CLI)

Create database

```
oci db database create --compartment-id <compartment_ocid> \
  --db-home-id <db_home_ocid> \
  --db-name <db_name> [OPTIONS]
```

Delete database

```
oci db database delete --database-id <database_ocid> [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_database" "cdb" {
  compartment_id = var.compartment_id
  db_home_id     = var.db_home_id
  display_name   = "<db_display_name>"
  db_name        = "<db_name>"
…
}
```

Local VM Command-Line Interfaces - dbaascli

Create a new database

```
$ dbaascli database create --dbName <value> {--oracleHome <value> | --oracleHomeName <value>}
  [--dbUniqueName <value>] [--dbSID <value>] [1]…
```

Create a new database from active database

```
$ dbaascli database duplicate --dbName <value> --sourceDBConnectionString <value>
  {--oracleHome <value> | --oracleHomeName <value>} [--dbSID <value>] [--dbUniqueName <value>] [1]…
```

Convert given non-CDB database to PDB

```
$ dbaascli database convertToPDB --dbname <value> {--copyDatafiles [--keepSourceDB] | --backupPrepared}
  [--cdbName <value>] [1]…
```

Create a new database template

```
$ dbaascli database createTemplate --dbname <value> {--templateLocation <value> | --uploadToObjectStorage
  --objectStorageLoginUser <value> --objectStorageBucketName <value> [--objectStorageUrl <value>]} [1]…
```

Delete the given database

```
$ dbaascli database delete --dbname <value> [--deleteArchiveLogs <value>] [--deleteBackups <value>] [1]…
```

[1]… Not all parameters listed. Please check the command definition

Provision Pluggable Databases

Provision a Pluggable Database

*A **Pluggable Database (PDB)** is a self-contained, portable collection of schemas and objects that operates within a multitenant Container Database (CDB).*

Requirements

- CDB already created
- Oracle Database 19c and later

Limitations

- ***Discovery:*** PDBs created via SQL aren't shown in OCI Console immediately; OCI syncs them automatically within ~45 minutes.
- ***Backups:*** Console/API backups occur at the CDB level and include all PDBs; dbaascli can back up individual PDBs.
- ***Restores:*** Console/API restores are CDB-level; dbaascli allows restoring specific PDBs.
- ***Data Guard:*** Creating a (PDB) using the console is not supported for databases using Data Guard.

Considerations

- During PDB creation the parent database (CDB) is in "Updating" state
- You can create up to 10 PDBs at the same time (parallel creation is allowed)
- PDB would be created in all the nodes of the VM Cluster.

How can you create a Pluggable Database?

You can create a Pluggable Database to an existing Container Database by using:

- OCI Console, OCI CLI, REST API, SDK, IaC
- dbaascli, SQL (not recommended)

OCI Control Plane Interfaces

OCI Console

Create — *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata VM Clusters ➔ Container databases ➔ Select CDB ➔ Pluggable Databases ➔* ***Create***

Delete — *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ Exadata VM Clusters ➔ Container databases ➔ Select CDB ➔ Pluggable Databases ➔ Select Actions (3 dots) on the PDB ➔* ***Delete***

Command Line Interface (OCI CLI)

Create PDB

```
oci db pluggable-database create -pdb-name <pdb_name> \
  --container-database-id <cdb_ocid> [OPTIONS]
```

Delete PDB

```
oci db pluggable-database delete \
  --pluggable-database-id <pdb_ocid> [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_pluggable_database" "test_pdb" {
    #Required
    container_database_id = oci_database_database.test_cdb.id
    pdb_name = var.pluggable_database_pdb_name
…
}
```

Local VM Command-Line Interfaces - dbaascli

Create a new PDB

`$ dbaascli pdb create --pdbName <value> --dbName <value>` [1]…

Delete a PDB

`$  dbaascli pdb delete --dbName <value> {--pdbName <value> | --pdbUID <value>}` [1]…

Clone a PDB from another PDB in the same database

`$ dbaascli pdb localClone --pdbName <value> --dbName <value> [--targetPDBName <value>]` [1]…

Clone the specified PDB from another database

`$ dbaascli pdb remoteClone --pdbName <value> --dbName <value> --sourceDBConnectionString <value>` [1]…

Relocate the specified PDB from remote DB into local DB

`$ dbaascli pdb relocate --pdbName <value> --dbName <value> --sourceDBConnectionString <value>` [1]…

Refresh a specified PDB

`$ dbaascli pdb refresh --dbname <value> {--pdbName <value> | --pdbUID <value>} [--waitForCompletion <value>]`

[1]… Not all parameters listed. Please check the command definition

Provision Data Guard Association

Provision a Data Guard Association

*A **Data Guard association** is the logical pairing between a primary database and its standby database that defines their relationship, replication mode, and failover/switchover capabilities.*

Requirements

- ***Infrastructure:***
  - Two separate Exadata VM Clusters
  - Same VCN and region (or VCN peering configured for cross-region)
  - Standby VM Cluster in a different availability domain (recommended)
- ***Database:***
  - Primary and standby databases must have the same version
  - Primary database must exist and be functioning
- ***Database Home:***
  - Create a new DB Home on standby instance OR use an existing one
  - Standby DB Home matching all patches matching on the primary
- ***Network:***
  - Listener port (TCP) open between primary and standby subnets
  - Ingress and egress security rules configured (stateful)
  - Data Guard replication occurs over the client network
- ***Encryption (if using customer-managed keys):***
  - Oracle Vault Service must be configured
  - Master encryption key must be created
  - Key version OCID must be available
- ***Passwords:***
  - SYS password required for primary database
  - TDE wallet password (if using Oracle-managed encryption)

Provision a Data Guard Association

Configurations & Types

- Physical Standby
- Up to 6 standby databases
- Data Guard Type:
  - Data Guard or Active Data Guard
- Protection Mode:
  - Maximum Availability (Sync) or Maximum Performance (Async)
- Cross-Service Data Guard:
  - Cross-service Data Guard between ExaDB-D and ExaDB-XS requires Oracle AI Database 26ai

Supported Operations

- ***Switchover:*** Reverses the primary and standby database roles.
- ***Failover:*** Transitions the standby database to the primary role after the primary fails or becomes unreachable.
- ***Reinstate:*** Reinstates a failed database into the standby role after correcting the cause of failure.
- **PDB creation:** Not supported using the console
- Physical ⇔ Snapshot conversion

Restrictions

- **You can't terminate a primary with an active standby**: delete the standby first or perform a switchover.
- **You can't terminate a VM cluster containing Data Guard enabled databases:** remove the DG association by deleting the standby first.

How can you create a Data Guard Association?

You can create a Data Guard Association to an existing Container Database by using:

- OCI Console, OCI CLI, REST API, SDK, IaC
- dbaascli, Data Guard Broker, SQL (not recommended)

OCI Control Plane Interfaces

OCI Console

Create *Menu ➔ Oracle AI Database ➔ Oracle Exadata Database Service on Dedicated Infrastructure ➔ VM Clusters ➔ Container databases ➔ Select CDB ➔ Data Guard Associations ➔* ***Add standby***

Command Line Interface (OCI CLI)

Create Data Guard Association

```
oci db data-guard-association create from-existing-vm-cluster \
  --database-admin-password <adm_pwd> \
  --database-id <db_ocid> \
  --peer-vm-cluster-id <cluster_ocid> \
  --protection-mode <protection_mode> \
  --transport-type <transport_type> [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_data_guard_association"
"test_dg_assoc" {
    #Required
    creation_type = var.data_guard_association_creation_type
    database_admin_password =
var.data_guard_association_database_admin_password
    database_id = oci_database_database.test_database.id
    delete_standby_db_home_on_delete =
var.data_guard_association_delete_standby_db_home_on_delete
    protection_mode =
var.data_guard_association_protection_mode
    transport_type =
var.data_guard_association_transport_type
...
}
```

To ***terminate*** *an Oracle Exadata Database Service on Dedicated Infrastructure instance Data Guard association, you* ***delete the standby database****.*

Local VM Command-Line Interfaces - dbaascli

Configure a Standby in an existing Data Guard setup

```
$ dbaascli dataguard configureStandby --dbname <value> --standbyDBUniqueName <value> [1]…
```

Configure an Oracle database as a Primary database

```
$ dbaascli dataguard prepareForStandby --dbname <value> --standbyDBUniqueName <value> [1]…
```

Export Primary wallet for Standy creation

```
$ dbaascli dataguard prepareStandbyBlob --dbname <value> --blobLocation <value>
```

Enable multiple Standby support on DG setup

```
$ dbaascli dataguard enableMultipleStandbySupport --dbname <value> [--executePrereqs] [--primaryDBOCID <value>] [1]…
```

Register Standby

```
$ dbaascli dataguard registerStandby --dbname <value> --standbyDBUniqueName <value> [1]…
```

Deregister Standby

```
$ dbaascli dataguard deregisterStandby --dbname <value> --standbyDBUniqueName <value> [1]…
```

[1]… Not all parameters listed. Please check the command definition

Stay tuned

Operations Advisory
GitHub Repository

Thank you

ORACLE